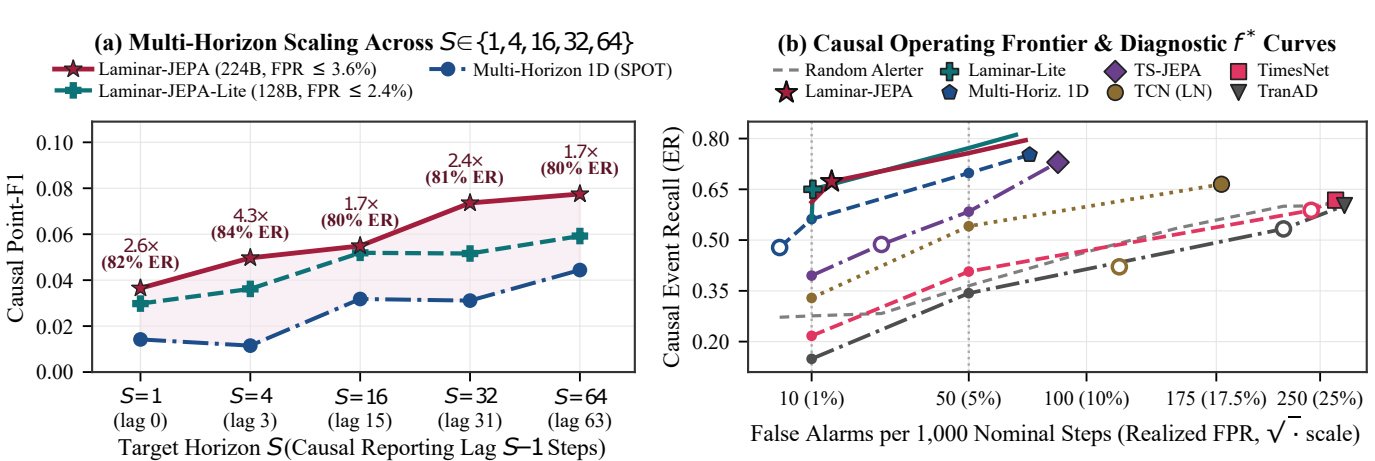

(a) Multi-Horizon Scaling Across $S \in \{1, 4, 16, 32, 64\}$
Laminar-JEPA (224B, FPR ≤ 3.6%)
Laminar-JEPA-Lite (128B, FPR ≤ 2.4%)
Multi-Horizon 1D (SPOT)
Causal Point-F1
0.10
0.08
0.06
0.04
0.02
0.00
2.6× (82% ER)
4.3× (84% ER)
1.7× (80% ER)
2.4× (81% ER)
1.7× (80% ER)
S=1 (lag 0)
S=4 (lag 3)
S=16 (lag 15)
S=32 (lag 31)
S=64 (lag 63)
Target Horizon S (Causal Reporting Lag S−1 Steps)
(b) Causal Operating Frontier & Diagnostic f^* Curves
Random Alerter
Laminar-JEPA
Laminar-Lite
Multi-Horiz. 1D
TS-JEPA
TCN (LN)
TimesNet
TranAD
Causal Event Recall (ER)
0.80
0.65
0.50
0.35
0.20
10 (1%)
50 (5%)
100 (10%)
175 (17.5%)
250 (25%)
False Alarms per 1,000 Nominal Steps (Realized FPR, $\sqrt{\cdot}$ scale)